



GAME DESIGN DOCUMENT

Build a factory. Arm it. Hold the line.

FUZENOVA
GAMES

PLAYABLE
PROTOTYPE

VERSION
0.1.0

2026-10-
07

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Everything numeric in this document (prices, stats, recipes, level scaling) is read directly from the game's code when the document is built, so it matches the game as it is today. Items marked **PLANNED** or **ART READY** are designed but not yet in the game.

01 Vision

Rogue machines march on your base. You survive by running a factory: mine ore, smelt it, craft ammunition and robots, wire up power, and hold the line through level after level of tougher waves. It is Factorio and Mindustry distilled into a short, replayable, browser-based run, with a burnt, ember-lit sci-fi look.

DESIGN PILLARS

A factory that fights

The satisfaction of an automated production line, but every belt, machine and stockpile exists to keep you alive. No sandbox sprawl: everything you build is judged by the next fight.

Short, complete runs

A level is a few minutes. A star is earned at level 10, 20 and 30. It is designed for a browser tab, not a 40-hour save file, with autosave so a run can be paused and resumed.

Readable at a glance

Clear silhouettes, blue-and-orange friendly machines against rusted red-glow enemies, and a UI that explains every building the first time you pick it up.

Forgiving, not shallow

Building is always allowed, a broken belt is cheap to fix, repairs between rounds are free. Depth comes from supply chains and choices, not punishment.

AT A GLANCE

Genre	Factory-building tower defence / roguelite
Platform	Web browser (installable as an app, works offline). Windows launcher planned.
Inspired by	Factorio, Mindustry, Satisfactory; the round structure of wave defence games
Session length	3 to 10 minutes per level, autosaved. A full 30-level clear is roughly 2 to 3 hours.
Camera / art	Top-down 2D, painted sprites, dusk lighting with glow effects
Team so far	One designer/producer plus AI-assisted code and art

02 The core loop

1

Build

Between fights there is a build phase (90 seconds by default, or start the fight early). The factory is paused. Place drills, belts, smelters, assemblers, turrets, walls, power and robot fabricators. Buildings are paid for from the core stockpile.

2

Fight

The factory runs while a fight is on, and for a breather after each win (50 seconds on Easy, 30 on Normal, 20 on Hard, 12 on Extreme at level 1, shrinking by 2 percent a level to 40 percent), so you can mine and smelt while you build. You can keep building during a fight too, so a jam is never fatal. Enemy waves spawn at the edge of the map and march on the core, attacking turrets and walls first.

3

Win or lose

Clear the wave before the core falls; the level ends the moment the last enemy dies. Lose the core and the run is over. Win and every surviving building is repaired (on Hard and Extreme it costs plates); robots stay wounded, and dead ones must be built again.

4

Pick a boon

After every second win (2, 4, 6...) and after each boss, choose one of three random boons: a permanent upgrade such as turret damage, coil damage or sturdier walls, or an instant reward such as plates, science packs, free scouts or a full repair. They are deliberately modest, so a run is shaped by the luck of the draw without being broken by it.

5

Open a square

The map is a grid of 24x24 squares and you start with only the one round your core, which holds iron and nothing else. After each win you choose one of three neighbouring squares to open, and the game opens a second, random one. Copper comes next door, then coal and tin, then lead and sulfur, then the far forests, and a message explains each new resource the first time it appears. The map, and the danger, grow with every level.

6

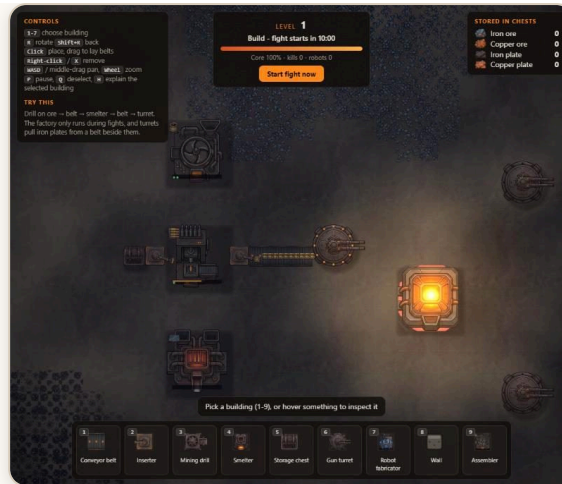
Read the omen

From level 2, a fight may carry an omen shown in the top bar before it starts: Swarm, Ironhide, Swift, Acid rain, Two fronts, Bounty, Brutal, Ash fog or A lull. Each run also has its own random map, so no two runs are alike.

7

Repeat, harder

The next level has more enemies, tougher enemies, more ranged attackers and new enemy types. Fights grow from 3 minutes toward 10.



An ammo press turning casings and gunpowder into bullets for a turret, with a gunpowder mill and alloy furnace beside it.

THE FACTORY IS PAUSED BETWEEN FIGHTS

Machines, belts and inserters only run while a fight is on. Between fights the base idles (belt arrows creep along at 35% speed, smelters puff smoke, robots patrol, turrets scan), but nothing is produced. That makes the build phase a planning phase, and makes the fight the test of the design.

03 Levels and pacing

One **level** is one build phase followed by one fight. The build phase is 90 seconds by default (the player may press *Start fight now* early). Fights get longer as you go: **3:00 at level 1, growing to 10:00 by level 30**, and staying at 10:00 beyond that.

Each fight spawns a fixed number of enemies across the first 85% of the timer, arriving in packs from one direction (pack size $1 + \text{level}/2$, up to 8), about 38 to 46 tiles from the core, and they march straight at it. The fight ends when the timer is out and every enemy is dead. Enemy strength scales per level; a basic crawler's stats look like this:

LEVEL	FIGHT LENGTH	ENEMIES	CRAWLER HEALTH	SPEED (TILES/S)	DAMAGE /S	NEW ENEMIES
1	3:00	20	36	0.9	2	Crawler
2	3:14	26	41	1	2.4	Swarmling
3	3:29	33	46	1	2.9	Armoured crawler
5	3:58	46	56	1.1	3.8	Scout drone
8	4:41	65	71	1.1	5.1	Siege crawler, Blade spider
10	5:10	78	81	1.2	6	
15	6:23	98	84	1.4	8.3	
20	7:35	117	86	1.5	10.5	
25	8:48	137	88	1.7	12.8	
30	10:00	156	90	1.8	15	

Formulas: enemies = $\text{round}(5 + 2.5 \times \text{level})$; health = $24 \times (1 + 0.2 \times \min(\text{level} - 1, 9) + 0.03 \times \max(0, \text{level} - 10))$; speed = $\min(2.2, 0.9 + 0.03 \times \text{level})$; damage per second = $2.5 + 0.75 \times \text{level}$. Each enemy type multiplies these (see Enemies).

These numbers are a first guess and untested past the early levels.

WHEN ENEMY TYPES ARRIVE

L1 Crawler L2 Swarmling L3 Armoured crawler L4 Spider-walker L4 Acid spitter L5 Scout drone L8 Siege crawler
L8 Blade spider L9 Acid sprayer L9 Combat drone L11 Gatling brute L12 Laser spider L13 Acid cannon
L14 Bomber drone L17 Siege spitter

STARS AND ENDLESS (PLANNED)

Each commander earns **one star at level 10, two at level 20, three at level 30**. After that, **Endless** keeps the levels coming at ever-increasing difficulty until you fall. The three star levels are natural places for boss fights (see Bosses).

04 Economy and allowances

Everything you build is paid for from the **Core stock**, a stockpile held by the Cinder Core. You begin with **170 iron plates and 45 copper plates**. There are four ways to top it up:

- **Deliver plates to the core.** A belt running into the core, or an inserter dropping into it, adds its items to the stock. This is the main income and needs a real supply line.
- **Kills.** Each enemy destroyed pays 2 iron plate.
- **Level bounty.** Winning a level pays 30 iron plate and 8 copper plate, growing 5% per level.
- **Removing buildings** refunds 75% of the cost. Anything placed during the current build phase refunds in full, because nothing is locked in until the fight starts. After that, belts and inserters still refund in full and other buildings refund 75%. Placing still checks that you can afford it, so the budget is reserved as you plan.

Turning an existing belt round is free. If the core cannot pay, the build bar dims the button and shows what is missing. The tutorial level is free to build in.

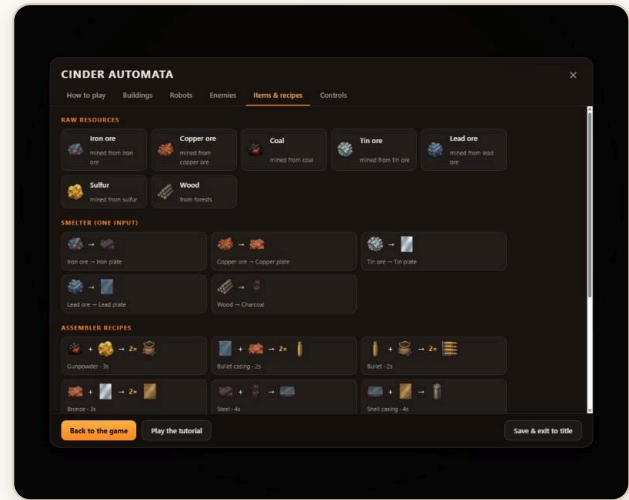
WHAT THINGS COST

BUILDING	SIZE (TILES)	HEALTH	COST
Conveyor belt	1×1	40	free
Inserter	1×1	40	free
Mining drill	3×2	140	16 iron
Smelter	2×2	140	10 iron
Wall	1×1	320	4 iron
Gun turret	2×2	280	30 iron + 5 copper
Assembler	3×3	160	40 iron + 10 copper
Drone workshop	3×3	240	60 iron + 20 copper
Gunship hangar	3×3	240	80 iron + 35 copper
Walker foundry	3×3	240	80 iron + 35 copper
Heavy works	3×3	320	130 iron + 60 copper
Power pole	1×1	140	2 iron + 1 copper
Ember generator	2×2	220	40 iron + 10 copper
Scrap bin	2×2	120	20 iron
Flamer	2×2	260	40 iron + 15 copper
Tunnel	1×1	80	8 iron
Storm coil	2×2	260	50 iron + 30 copper

BUILDING	SIZE (TILES)	HEALTH	COST
Crossover	1×1	40	free
Splitter / merger	1×2	60	free



The core stock panel (top right) and prices on the build bar.
Unaffordable buildings dim.



The in-game Items & recipes page.

OTHER LIMITS

- Turrets keep an ammo stock of up to 40 shots. Generators hold up to 120 seconds of fuel.
- Each fabricator has room for **12 space** of robots. A small scout or drone takes 1, a Trooper or Gunship 2, a Heavy walker 4, a Turret walker 5, Mobile artillery 6, and a Titan takes all 12. When its robots die, room frees up and it builds more. A gauge above each fabricator shows it.
- **Ore is finite.** Each tile holds about 300 to 1500 units and depletes as drills mine it. New ore comes from the resource modules you place after each win.
- Resource modules: choose 1 of 3 (a different mix each level), then click the map to stamp a rich patch (radius 4.2 tiles, 1.5× richness). It never lands on top of buildings.

05 Buildings

Twelve buildings plus the core. All are placed on a tile grid; most can be dragged out in lines. The first time a building is picked up, the game explains what it is, how to connect it and what it does (also in the in-game menu).



Conveyor belt

1×1 40 health free

A **conveyor belt** carries items along, one tile at a time, in the direction of its arrows.

Moves ore and plates around your base. Turrets and robot buildings grab items off any belt that touches them.

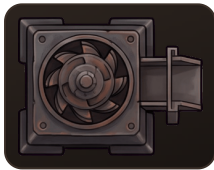


Inserter

1×1 40 health free

An **inserter** is a robotic arm that moves one item at a time from one thing to another.

Loads a smelter from a belt, and unloads the finished plates from the smelter onto another belt. **Double-click an inserter** (with no building selected) to make it move only one kind of item, for example copper ore off a mixed belt; leave it on "Any item" to move everything.



Mining drill

3×2 140 health 16 iron

A **mining drill** digs ore out of the ground. It is 3 tiles long and 2 wide, and only works on top of ore (the more ore under it, the better the spot).

Produces ore non-stop while a fight is on. Each ore tile runs out eventually, so ore patches thin out.



Smelter

2×2 140 health 10 iron

A **smelter** melts raw ore into metal plates (iron ore into iron plates, copper ore into copper plates). It is 2x2 tiles.

Turns ore into plates, one every few seconds. Plates are what turrets and robots use as ammo and parts.



Assembler

3×3 160 health

40 iron + 10 copper



An **assembler** combines several materials into something new: gunpowder, bullets, bronze, steel and artillery shells. It is 3x3 tiles.

Recipes: coal + sulfur = gunpowder; lead + copper = bullet casing; casing + gunpowder = bullets; copper + tin = bronze; iron + charcoal = steel; steel + bronze = shell casing; casing + 2 gunpowder = artillery shell. (Make charcoal by smelting wood.)



Drone workshop

3×3 240 health

60 iron + 20 copper

The **drone workshop** builds the small robots: Scout drones and Scout walkers. It is 3x3 tiles, and the first of four robot buildings.

While a fight is on it builds robots that walk or fly out on their own, shoot enemies and do not heal once hurt. It holds 12 space of robots at once. Bars under it show build progress (blue) and plates loaded (gold).



Wall

1×1

320 health

4 iron



A **wall** is a tough one-tile block that does nothing except take hits.

Enemies attack turrets and walls before other buildings, so walls soak up damage while turrets shoot.

Damaged buildings are repaired for free between levels; destroyed ones have to be placed again.



Power pole

1×1

140 health

2 iron + 1 copper

A **power pole** carries electricity.

Poles link to each other and power any generator or coil close to them.

Nothing on its own; it joins generators and coils into one network so they share power.



Ember generator

2×2

220 health

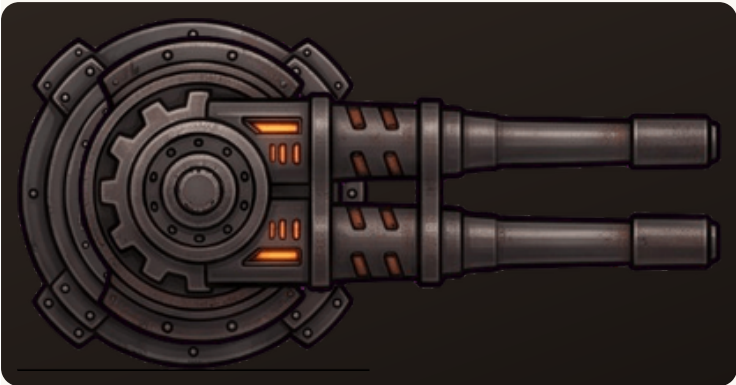
40 iron + 10 copper

An **Ember generator** burns fuel to make power. It is 2x2 tiles.

Makes 100 power while it has fuel and a fight is on. Each Storm coil in action uses about 70, so one generator runs about one coil at full strength. Too little power and the coils hit weaker.

Self-orienting parts: **drills** hand ore to any belt or machine touching them on any side and turn to face it; **inserters** turn themselves to face the direction items can flow, then lock after their first delivery so they can never pull finished goods back the wrong way.

06 Weapons and defences



Gun turret



Storm coil



Walls

GUN TURRET

Range 7 tiles, one shot every 0.42 seconds. Fed by any belt that touches it (no inserter needed), it keeps a stockpile of 40 shots; ring of dots around its rim shows how full it is. Damage depends on the ammunition last loaded:

AMMO	SHOTS PER ITEM	DAMAGE PER SHOT	DAMAGE PER SECOND
Iron plate	4	8	19
Bullet	10	20	47.6
Artillery shell	3	70	166.7
Coal	20	6	14.3
Charcoal	26	8	19
Wood	10	4	9.5

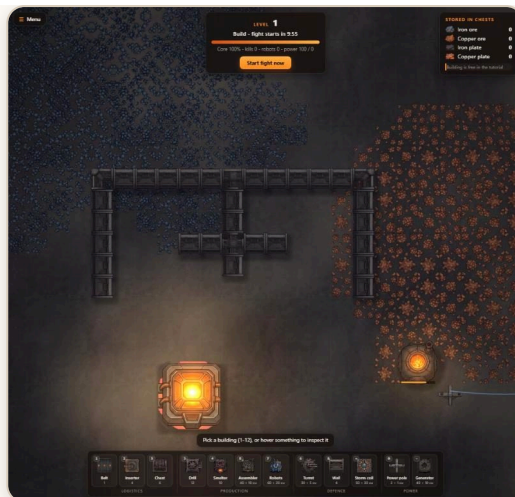
A turret loaded by hand-placed test ammo does 8 damage per shot. Planned turret types (art ready): flame, laser, rocket and artillery.

STORM COIL

Needs **no ammunition** but does need power. Every 1.3 seconds it fires a bolt of chain lightning at the nearest enemy within 6.5 tiles for 30 damage, then jumps to the nearest un-hit enemy within 3.4 tiles, up to 2 more times, each hop hitting 75% as hard as the last. A full four-target volley deals about **69 damage** in total. It stores energy: it charges up to 200 at 60 power, each bolt spends 50, and a full coil draws only 2. So a charged coil fires a burst of 4 bolts and then only as fast as the network can recharge it. A network short of power recharges slowly; with no power, it never charges. It is the best answer to crowds of weak enemies.



A Storm coil chaining lightning through four enemies.



Self-connecting walls (corners, a T and a cross) and a powered generator and pole.

WALLS

320 health, dragged out in lines. They join up automatically into straights, corners, T-junctions and crosses. Enemies attack turrets, walls and coils first, so walls buy time.

POWER

- **Ember generator:** burns fuel (coal 25s, charcoal 30s, wood 12s) and gives 100 power while a fight is on.
- **Power pole:** links to other poles within 7.5 tiles, powers anything within 3.5 tiles. Each connected group of poles is one network; machines share its power, and a network short on supply scales everything down together.

07 Research

Everything you build starts deliberately modest. **Research** (press T) is how it grows, and it is paid for in **science packs** that you have to craft: set an Assembler to a science pack recipe, feed it plates and belt the packs into the core. There are three kinds (projectile: iron + copper; electromagnetic: copper + tin; robotics: iron + lead), so each branch needs its own production line and its own ore. Some research makes things better; some **unlocks** things: robot designs (Trooper and Gunship drone, Heavy walker, Turret walker, Mobile artillery, Titan) and ammunition (bullets, then steel and artillery shells). A new game starts with only the smallest robots and no bullets.

BRANCH	RESEARCH	EFFECT PER LEVEL	LEVELS	COST OF THE FIRST / LAST LEVEL
Projectile weapons	Projectile damage	+12% damage per shot from gun turrets	5	5 projectile packs / 26 projectile packs + 13 em packs + 11 advanced packs
Projectile weapons	Rate of fire	Gun turrets fire 6% faster	5	4 projectile packs / 24 projectile packs + 12 em packs + 10 advanced packs
Projectile weapons	Targeting optics	+10% range for gun turrets	3	6 projectile packs / 16 projectile packs + 8 em packs + 7 advanced packs
Projectile weapons	Ammunition	Level 1 unlocks bullets (casings and bullets in the Assembler). Level 2 unlocks steel and artillery shells.	2	8 projectile packs / 16 projectile packs + 8 em packs
Projectile weapons	Turret designs	Level 1 unlocks the Scatter gun (short range, hits everything in a cone). Level 2 unlocks the Sniper (long range, slow, hits very hard). Level 3 unlocks Artillery (a very long range, bursts over an area, needs shells; upgrade a Scatter gun or Sniper into it). Upgrade a gun turret by double-clicking it.	3	6 projectile packs / 20 projectile packs + 10 em packs + 8 advanced packs

BRANCH	RESEARCH	EFFECT PER LEVEL	LEVELS	COST OF THE FIRST / LAST LEVEL
Projectile weapons	Flame designs	Level 1 unlocks the Flamer (build it from the Defence group; it burns coal, charcoal or wood). Level 2 unlocks the Incendiary launcher and the Focused torch. Level 3 unlocks the Plasma cannon. Upgrade a Flamer by double-clicking it.	3	6 projectile packs / 18 projectile packs + 9 em packs + 8 advanced packs
Fortification	Reinforced structures	+15% health for turrets and walls	4	6 projectile packs / 24 projectile packs + 12 em packs + 10 advanced packs
Electromagnetic weapons	Electromagnetic damage	+15% lightning damage from Storm coils	5	5 em packs / 30 em packs + 15 robotics packs + 12 advanced packs
Electromagnetic weapons	Coil designs	Level 1 unlocks the Shield coil (protects nearby buildings) and the Stun coil (freezes enemies for a moment). Level 2 unlocks the Railgun (a long piercing shot). Upgrade a Storm coil by double-clicking it.	2	6 em packs / 12 em packs + 6 projectile packs
Electromagnetic weapons	Chain conductors needs Electromagnetic damage 2	Lightning jumps to 1 more enemy	3	8 em packs / 22 em packs + 11 robotics packs + 9 advanced packs
Electromagnetic weapons	Field focusing	+10% range for Storm coils	3	6 em packs / 16 em packs + 8 robotics packs + 7 advanced packs
Robotics: designs and ammunition	Robot designs	Unlocks new robots: 1 Trooper and Gunship drone, 2 Heavy walker and Sapper drone, 3 Turret walker, 4 Mobile artillery, 5 Titan and Carrier	5	6 robotics packs / 36 robotics packs + 18 projectile packs + 15 advanced packs
Robotics: designs and ammunition	Robot ammunition	Better rounds for your robots: +12% robot damage (hardened, piercing, incendiary, plasma)	4	6 robotics packs / 24 robotics packs + 12 projectile packs + 10 advanced packs
Robotics: designs and ammunition	Robot plating	+12% robot health	4	6 robotics packs / 24 robotics packs + 12 projectile packs + 10 advanced packs

BRANCH	RESEARCH	EFFECT PER LEVEL	LEVELS	COST OF THE FIRST / LAST LEVEL
Robotics: designs and ammunition	Robot sights	+10% weapon range for your robots	3	8 robotics packs / 22 robotics packs + 11 projectile packs + 9 advanced packs
Research applies at once, including to buildings already standing. The design rule: base weapons stay modest so research and commanders have room to matter; a fully researched turret fires roughly 2.3 times as much damage as a fresh one. Since robots do not heal, the robotics line also decides how well your army lasts. In the headless test a typical player with the projectile techs at level 2 to 5 needs about half as many turrets as one with none. More branches are planned: logistics (faster belts and drills), artillery and flame weapons, and robot tiers. The Forewoman commander is meant to research faster, which is not built yet.				

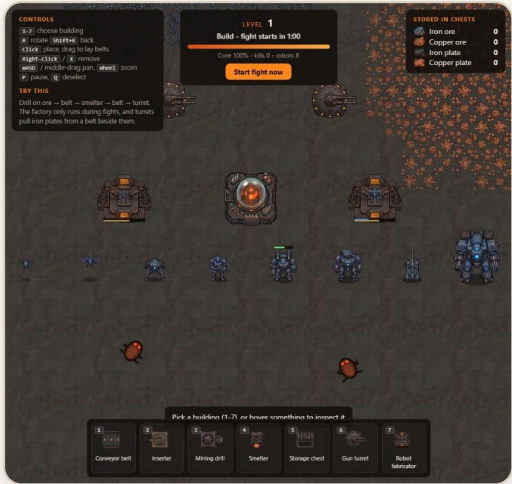
08 Allies: robot soldiers

Your soldiers are robots built in a **Robot fabricator** from iron plates. Click a fabricator to choose which robot it builds. While a fight is on, each fabricator pulls plates from an adjacent belt and rolls out a robot when it has enough. Robots then act on their own:

- They walk toward enemies within about 16 tiles, stop at their weapon range and shoot, turning to face their target.
- With no enemies around, each stands at its own guard post just outside the base, facing outward, and they keep out of each other's way.
- Ground crawlers stop to bite ground robots in their way, but **cannot reach flying ones**.
- They keep going from level to level, but **do not heal**: a wounded robot stays wounded, and dead ones are replaced by building new ones. The size of the army is limited by the room in your fabricators.

ROBOT	COST	BUILD TIME	HEALTH	RANGE	DAMAGE /S	SPEED
 Scout drone flying	[object Object] iron	6s	18	4.5	6	3
 Scout walker	[object Object] iron	9s	55	4	10	2.4
 Gunship drone flying	[object Object] iron	9s	30	5	14	2.8
 Sapper drone flying	[object Object] iron	12s	70	0.9	34	2.4
 Trooper	[object Object] iron	12s	75	5.5	16	1.9
 Turret walker	[object Object] iron	20s	240	6.5	40	1.3
 Heavy walker	[object Object] iron	18s	190	5	34	1.4
 Mobile artillery	[object Object] iron	24s	130	10	22	1.1
 Titan	[object Object] iron	40s	720	6	110	1

ROBOT	COST	BUILD TIME	HEALTH	RANGE	DAMAGE /S	SPEED
 Carrier flying	[object Object] iron	45s	900	6	30	1



The full roster in front of two robot fabricators: scout drone to Titan. (Since this screenshot the robots were redrawn top-down so they turn to face targets.)

Friendly robots are drawn clean, dark steel with **blue and orange trim** and a glowing blue core, so they are instantly readable against rusted, red-glow enemies.

09 Enemies

The enemy is a swarm of **rogue, corroded automata**: rusted, sooty and glowing red. They spawn at the map edge and march straight at the core. On the way they attack, in order of priority: any ground robot they are touching; then the nearest **turret, wall or Storm coil** within 7 tiles; then other machines they are practically touching; otherwise the core. They ignore belts and inserters, so a stray belt can never ruin a run.

	ENEMY	FIRST SEEN	SPAWN WEIGHT	HEALTH	SPEED	DAMAGE	HEALTH AT L10	AT L30
	Crawler	1	10	×1	×1	×1	81	90
	Swarmling	2	3	×0.4	×1.8	×0.5	33	36
	Armoured crawler	3	5	×2.2	×0.9	×1.6	179	198
	Spider-walker	4	4	×1.6	×1.3	×1.2	130	144
	Acid spitter	4	6	×0.9	×1	×0.9	73	81
	Scout drone flying	5	4	×0.8	×1.6	×0.7	65	72
	Siege crawler	8	3	×5	×0.75	×3	407	450
	Blade spider	8	3	×2.4	×1.5	×2	195	216
	Acid sprayer	9	4	×1.6	×0.95	×1.3	130	144
	Combat drone flying	9	4	×1.8	×1.4	×1.5	146	162
	Gatling brute	11	3	×6	×0.7	×1.6	488	540

	ENEMY	FIRST SEEN	SPAWN WEIGHT	HEALTH	SPEED	DAMAGE	HEALTH AT L10	AT L30
	Laser spider	12	2	×6	×0.9	×3.5	488	540
	Acid cannon	13	3	×2.5	×0.85	×2.2	203	225
	Bomber drone flying	14	2	×5	×1	×3	407	450
	Siege spitter	17	2	×3.5	×0.8	×3	285	315

Multipliers apply to the level's base stats (see Levels and pacing). Spawn weight is how common the type is once unlocked.



Every enemy type at once, attacking the base: rusted machines against the friendly blue-trimmed robots and turrets.

MORE ENEMY TYPES (ART GENERATED, NOT YET IN THE GAME)



Brute (3 tiers) art ready, not built

Heavy armoured ground unit. Slow, very tough, hits hard against structures.



Acid spitter (4 sizes) art ready, not built

Ranged. Lobs acid at buildings from a distance, so it forces you to kill it before it is in range.

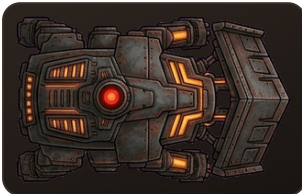


Heavy bomber drone art ready, not built

Flying. Drops bombs on buildings and cannot be blocked by walls. Only turrets, coils and drones can reach it.

10 Bosses

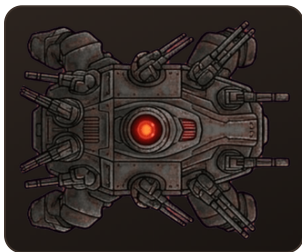
Every tenth level has a boss. It walks in 30% of the way through the fight, the level cannot end until it is dead, and it pays out an extra 150 iron and 50 copper (a bit more on higher levels). The order repeats: Brute, Colossus, Queen, Brute again at level 40, and so on into Endless.



Siege Brute

Level 10

A wall-breaker with a huge ram plate. Walks straight at the core, ignoring turrets that are not in its way, and smashes buildings in its path for 2.5 times the usual damage. The first-star test: it forces real firepower rather than a trickle of fire.



Colossus Walker

Level 20

A gigantic multi-turret walker. It opens fire on turrets, coils and walls from 9 tiles, beyond the gun turret range of 7, and keeps closing to about 6 tiles while it shells. Its shells hit hard, so it punishes a defence that is only a few turrets and rewards robots and heavy firepower. It does not attack robots.



Hive Queen

Level 30

The nest mother. She walks in, parks about 9 tiles from the core and does not attack directly. Every 6 seconds she spawns two swarmlings and a scout drone (up to 16 alive), so the fight is about killing her fast while holding off the brood. The final exam for a balanced defence, and the gate to Endless mode.

Design intent: each boss tests a different weakness (no real firepower, no answer to range, no answer to a stream of small enemies) and arrives on top of the normal wave. In our headless test a boss level asks for roughly twice the defence of the level before it. The star for the commander comes with commanders, which are not built yet. **Boss health and damage are first numbers, not tested by real play.**

11 Resources and the production chain

Seven raw resources sit in patches across the 160×160 map at a fair distance from the core, so getting them means planning a supply line. Smelters turn ore into plates, and assemblers combine several inputs into the ammunition and alloys the defence needs.



Iron ore



Copper ore



Coal



Tin ore



Lead ore



Sulfur



Forest

SMELTER (ONE INPUT)



Iron plate smelter



Copper plate smelter



Tin plate smelter

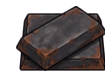


Lead plate smelter



Charcoal smelter

ASSEMBLER (SEVERAL INPUTS)



Projectile science pack 4s

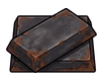


Electromagnetic science pack 5s

2x



Robotics science pack 6s



Advanced science pack 9s



Gunpowder 3s



Bullet casing 2s



Bullet 2s



Bronze 3s



Steel 4s



Shell casing 4s



Artillery shell 5s

Bullets need lead, copper, coal and sulfur, so the first ammo chain pulls from four different patches. Artillery shells additionally need tin, iron and wood (for charcoal). Shells and artillery turrets are part of the planned content.

12 Commanders and meta-progression

Built: choosing a commander, their bonuses and drawbacks, stars, and going on past level 30 (Endless).

Planned: Salvage, the Workshop, the codex and captured tiles.

- **Stars per commander.** One star for completing level 10, two for level 20, three (the maximum) for level 30.
- **Endless mode.** Keep playing at ever-increasing difficulty until you die. Proposed: unlocks at three stars, and records your best level per commander.
- **Salvage and the Workshop.** A permanent currency earned every run (win or lose) from levels reached, spawners captured and bosses beaten. It is spent on *sideways* unlocks: new buildings and robots added to the research pool, opening kits, 2 or 3 optional perks per commander. Small capped power upgrades only, so level 30 stays a real test.
- **Codex and badges.** Enemies, recipes and items you have met are logged, with badge icons for milestones.
- **Captured tiles.** Robots attack enemy spawner nests. Destroying one captures its tiles, which become buildable land.

COMMANDERS

The player picks a commander at the start of every new game. Each has a theme, a bonus and a drawback, so the choice changes how you build. Everyone starts with Captain Wren Halloway (standard issue: no bonus and no drawback, so there is no choice at first); the six others unlock through achievements, each tied to the way that commander plays; the last two wait for weapons that do not exist yet (artillery and flame turrets). Stars are kept per commander in the browser: one for completing level 10, two for level 20, three for level 30. The game does not stop at 30; from there it is Endless, and the best level reached is shown on the commander card. Names are original placeholders that avoid film and comic characters; they need a trademark check before release.

COMMANDER	THEME	BONUS	DRAWBACK
Captain Wren Halloway	Standard issue	No bonus	No drawback
Gunnery Chief Brakka Vesh	Arsenal	Turrets hit 30% harder and hold 50% more ammo	Robots have 25% less health and hit 20% softer
The Foundry Regent	Robot army	Robots build 30% faster and have 30% more health	Turrets hit 20% softer
Stormwarden Ilka Thorne	Lightning	Storm coils hit 40% harder and chain 2 extra jumps	Coils draw 50% more power and turrets hit 20% softer
Forewoman Tamsin Brassgate	Pure factory	Starts with 50% more plates and is offered a fourth map square to choose from	Turrets hit 10% softer and robots 10% less health
Doctor Pim Quillfeather	Drone swarm	Small robots (drones, scouts, troopers) build in half the time	Big robots take 50% longer to build
Bastion Grimwald Oaksworn	Fortress	Turrets and walls have 50% more health	Robots walk 25% slower

COMMANDER	THEME	BONUS	DRAWBACK
Marshal Ozric Bellwether	Artillery	Artillery, incendiary launchers and plasma burst 30% wider and hit 25% harder	All turrets hit 10% softer, so the guns do little against fast things up close
Cinder Queen Ysolde Ash	Fire	Flamers, torches, launchers and plasma hit 50% harder, and their fire burns harder too	Storm coils hit 20% softer, and every turret 10% softer

13 Strategy and player experience

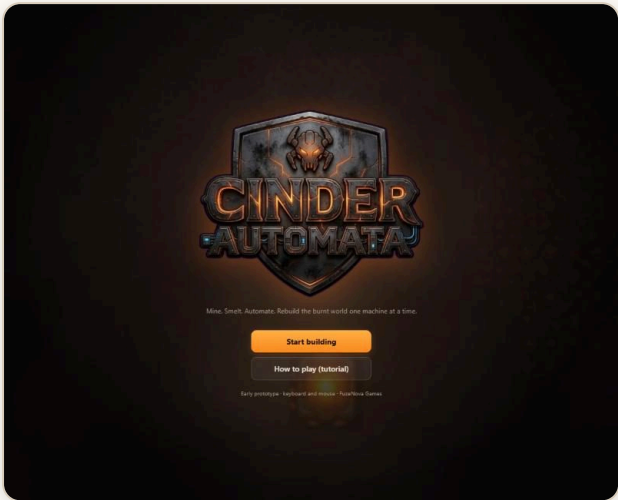
What a good run looks like, and what the game is trying to teach:

1. Early on, iron plates in a turret are enough. Build a second supply line before you need it.
2. Bullets (casing plus gunpowder) hit 2.5 times as hard as plates, so the ammo chain is worth building by level 5 or 6.
3. Enemies attack turrets, walls and Storm coils before anything else. Put a wall line in front of your turrets and they will spend their time chewing it.
4. Storm coils shine against crowds of weak enemies, and cost no ammo, but a coil with no fuel behind it does nothing. Keep the generator fed.
5. Robots fight between rounds too and are repaired for free. A steady trickle of cheap drones and troopers beats saving up for a Titan.
6. Run a belt of plates into the core to top up your build budget. Kills and finished levels also pay out.
7. Ore runs out. When you choose which square to open, pick the one holding the resource you are short of, not the prettiest one. Point at a square to see it on the map.
8. Ranged enemies (acid spitters, sprayers, cannons) stop short and shoot. Put walls in front, keep robots near to draw fire, and do not let a cannon sit out of reach of your turrets.
9. Read the omen before you press start: Two fronts means you need turrets on opposite sides; Ash fog shortens your turrets, so pull them in.
10. Fly a few drones: flying robots cannot be bitten by ground crawlers.

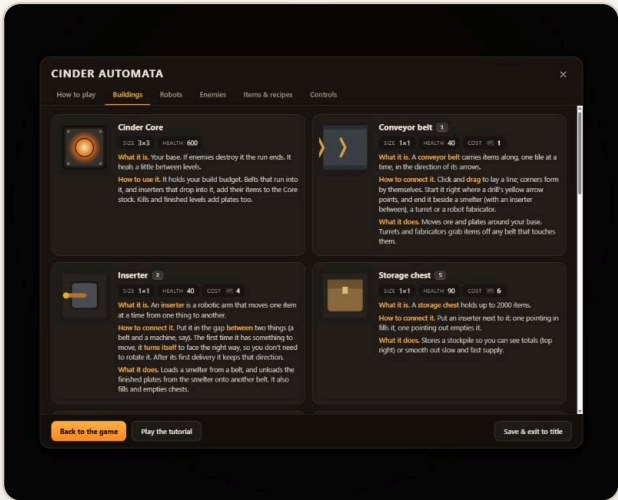
A TYPICAL EARLY RUN

1. **Levels 1 to 3.** Drill iron, smelt to plates, feed two or three turrets with a belt. Add a wall line. Enemies are basic crawlers.
2. **Levels 4 to 7.** Spiders and scout drones arrive. Add copper, a Robot fabricator and a few cheap robots. Start the ammo chain (coal, sulfur, lead) and switch turrets to bullets.
3. **Levels 8 to 12.** Armoured and siege crawlers, combat drones. Stand up power: a generator, poles and one or two Storm coils behind walls. Take resource modules that add what you are missing.
4. **Level 10 and beyond.** Waves of dozens of tough enemies, the first boss. A balanced defence with walls, turrets on bullets, coils and a steady stream of robots.

14 Interface and controls



Title screen with the game logo.



The in-game menu: every building with a picture, its size, health, cost and how to connect it.

- **Build bar** along the bottom, in groups (Logistics, Production, Defence, Power). Hover for a tooltip with the price; click for an explanation of how to connect it.
- **Round panel** top centre: level, phase timer, core health, kills, robots, power.
- **Core stock** top right. **Menu** top left: guide, and a picture book of every building, robot, enemy, item and recipe.
- A **guided tutorial** level walks a new player through mining, belts, smelting, feeding a turret and surviving a first fight.
- The interface scales smoothly with the window size.

Click / drag	Place the selected building; drag to lay belts and walls
Right-click or X	Remove (free until the fight starts, then 75%)
1 to 9, 0, -, =	Choose a building
R / Shift+R	Rotate
W A S D, wheel	Move and zoom the camera
H, P, Esc / M	Explain building, pause, menu

15 Art and audio direction

Look. A burnt, ash-covered alien world rebuilt with rugged industrial machines. Hand-painted top-down sprites with thick dark outlines, a muted charcoal, rust and soot palette, and warm ember-orange and amber glow as the accent. A cool dusk grade darkens the scene, and machines, the core, smelters and explosions cast real light onto the ground.

Colour language. Friendly: clean dark steel with blue and orange trim. Enemy: rusted, sooty, red glows. Power: cold blue-white. Fire and forges: orange.

How the art is made. Sprite sheets are generated in Google Gemini using a shared style block and reference images, then cut out of a flat magenta background, named and compressed by a script in the project. There are about 352 sprites so far. A prompt pack with copy buttons tracks what has been generated and what is still needed. Effects (explosions, smoke, lightning) are painted sprites blended with light; lightning bolts themselves are generated in code so they crackle differently every time.

Audio. None yet. The FuzeNova Games studio intro (an animated logo with a sound sting) plays before the game starts.

16 Technology and how it is built

Language / build	TypeScript, built with Vite. The simulation has no dependence on the screen, so it runs headless.
Rendering	HTML canvas, 2D, with a per-frame lighting pass. Holds 60 frames per second with 40 enemies, 25 robots and 40 explosions on screen.
Testing	Automated tests cover the factory, ammo chain, power, combat, damage, saving and offline behaviour.
Saving	Stored in the player's browser at the start of each build phase; restored exactly.
Distribution	Static files hosted on the site at /play/cinder-automata/, installable as an app (PWA) with offline play. A Windows launcher is planned.
Tools in the repo	Sprite slicer, prompt-pack builder, icon and logo tools, and this document's generator.

Status of the code: nothing has been committed to source control or deployed yet at the time of writing. The project lives in the **games/cinder-automata** folder of the luckyfuzion site repository, and builds into **public/play/cinder-automata**.

17 Where the project stands

CORE LOOP

- BUILT** Build phase, fight phase, win and lose, next level
- BUILT** Factory runs only during fights; building always allowed
- BUILT** Map expansion: choose one of three squares after each win, the game opens a second (*replaces the resource-module pick; 7x7 grid of 24x24 squares*)
- BUILT** Resources introduced by distance (iron, then copper, coal and tin, then lead and sulfur, forests) (*with a message the first time each appears*)
- BUILT** Random map seed per run; ?seed=N to replay
- BUILT** Omens: a random twist on each fight (9 kinds)
- BUILT** Boons: pick 1 of 3 random upgrades every second level (*modest by design; needs play-testing*)
- BUILT** Pause overlay; nothing can be built or changed while paused
- BUILT** Autosave and Continue (*saved at the start of each build phase*)
- BUILT** Email and password accounts with cloud saves (Firebase) (*needs the Firestore rules published before going live*)
- BUILT** Guided tutorial level

FACTORY

- BUILT** Drills, belts (with corners), inserters with filters, smelters
- BUILT** Assemblers with 7 recipes (ammo, alloys, shells)
- BUILT** Self-orienting drills and inserters
- BUILT** Building costs, core stockpile, refunds
- BUILT** Crossovers (belts crossing) and splitter / merger pieces (*drawn in code for now; art prompt to do*)
- ART READY** Two-lane belts, underground belts (*art generated, not built*)
- BUILT** Minimap, range circles, inspect mode
- BUILT** Small starting map that grows per level; rocks, burnt trees, tar pits, ridges (*obstacles drawn in code for now; decoration art to come*)
- BUILT** Research (T): paid in science packs you craft; projectile, electromagnetic, robotics, fortification, range (*unlocks robot designs and ammunition; more branches planned*)
- BUILT** Move buildings (V), controls panel (F2), inspect mode (F3), warning marks, fast-forward (F)
- BUILT** Blueprint planning: free to change until the fight starts
- BUILT** Storage chests (*removed on purpose; old saves move their contents to the core*)

DEFENCE

- BUILT** Gun turret (plates or bullets)
- BUILT** Walls that connect themselves
- BUILT** Storm coil with power (generators, poles)
- BUILT** Storm coils charge up (200) and spend energy per bolt
- ART READY** Flame, laser, rocket and artillery turrets (*art generated, not built*)
- ART READY** Gates, radar, shield projector

ALLIES

- BUILT** 10 robot types built in four buildings (Drone workshop, Gunship hangar, Walker foundry, Heavy works; 12 space each), auto-defending
- BUILT** Smart robots: guard posts, spacing, path-finding round buildings, unstick
- BUILT** Robots rotate to face targets, patrol between fights

ENEMIES

- BUILT** About 16 enemy types unlocking through the levels, with enemies that path round obstacles
- BUILT** Natural obstacles: rocks, burnt trees, tar pits, ridges
- BUILT** Enemies attack turrets and walls first
- BUILT** Boss enemies at levels 10, 20, 30 (Brute, Colossus, Hive Queen) (*repeat every 10 levels*)
- BUILT** Ranged enemies: acid spitters, sprayers, cannons, siege spitters, gatling brutes, combat drones, laser spiders (*some out-range a gun turret; needs balance testing*)
- ART READY** Bomber drones as a dedicated building-killer
- ART READY** Enemy nests to capture (spawners)

META

- BUILT** Commander select as a carousel with portraits; stars at levels 10 / 20 / 30; Endless past level 30 (*Artillery and Fire commanders are locked until their weapons exist*)
- BUILT** Starter commander plus achievement unlocks; four difficulty levels
- PLANNED** Salvage, Workshop, Codex, badges

PRESENTATION

- BUILT** Painted art, dusk lighting, effects, in-game menu with codex
- BUILT** Installable web app with offline play
- BUILT** Sound effects, machine loops and music (menu, build, fight 1, boss, victory, defeat) (*fight 2 and 3 music and the combat / boss effects still to come*)
- BUILT** New-game screen, title screen and pause overlay restyled (*the in-game panels and result screens are next*)
- PLANNED** Balance tuning past level 7 (*the headless test cannot win from level 8 with turrets alone since ranged enemies and tougher health were added*)
- BUILT** Tester launcher (Windows zip) (*must be rebuilt (npm run tester) before sending*)

ROADMAP

Right now

Balance pass for levels 8 to 30. Ranged enemies, tougher health, omens and boons all landed together, and the headless turret-only test can no longer win from level 8. Re-measure with robots and coils, tune ranged counts, acid cannon and siege spitter range, and boon strength. Then play-test the first 10 levels on the small 24x24 start square to see whether it needs a bigger first square.

Next

Finish the graphics pass: science pack icons, crossover and splitter art, wall damage states, decoration. Restyle the in-game panels (top bar, toolbar, research, result and menu screens) to match the new title and new-game screens.

Then

Audio to finish: fight 2 and fight 3 music, combat and boss sound effects (shots, deaths, coil zaps), close sounds. Capturable enemy nests, more research branches (logistics, artillery, flame), more turret types (flame, laser, rocket, artillery) that unlock the Artillery and Fire commanders.

Then

Workshop and Salvage (permanent sideways unlocks), Codex and badges, wider commander bonuses now that modules are gone, more omens and boons, the alien-invasion story direction if the producer confirms it.

Release prep

Rebuild and test the tester pack, Windows launcher, itch.io page, trademark checks on the title and names, analytics, and an optional global leaderboard.

18 Open questions and help wanted

OPEN DESIGN QUESTIONS

DECISIONS SO FAR

TOPIC	DECISION	STATUS
Cost of building	Everything placed during a build phase is a free-to-change blueprint. It is paid for as you plan (the budget is reserved and you cannot overspend) and refunded in full if you remove or move it. It locks in when the fight starts; after that, belts and inserters are always free, and other buildings refund 75% when removed.	BUILT
Structure damage	Repairs between levels are free on Easy and Normal. On Hard they cost a quarter of the damaged building price (in proportion to the damage), on Extreme half, paid from the core stock; anything you cannot afford stays damaged.	BUILT
Difficulty levels	Four levels chosen on the new-game screen. Easy: no build timer, the player starts each fight, enemies 15% softer. Normal: 90 s. Hard: 70 s, enemies 15% health and 10% damage up, paid repairs. Extreme: 45 s, enemies 30% health and 25% damage up, costly repairs.	BUILT
Commander unlocks	Everyone starts with Captain Wren Halloway: standard issue, no bonus and no drawback, and no choice to make at first. The other commanders are earned by playing in their style: Brakka (8 gun turrets standing at once), the Foundry Regent (3 Robot fabricators), Tamsin (6 drills and 4 smelters), Ilka (5 Storm coils), Pim (10 robots alive) and Grimwald (20 walls). Once a second commander is unlocked, the new-game screen lets you choose. The Artillery and Fire commanders wait for their weapons. Unlocks are kept in the browser.	BUILT
Roguelite layer	Every run has a random seed (its own map), each fight from level 2 may carry an omen, and every second win offers a pick of one of three random boons. Boons are small on purpose (5 to 7 percent steps, small windfalls) so they add variety without letting the player outscale the waves. Starting stock (170 iron, 45 copper) and level rewards (30 iron, 8 copper) were cut to slow the economy.	BUILT
Map expansion	The map is a 7x7 grid of 24x24 squares. A run starts on the core square with iron only. After each win the player picks one of three neighbouring squares (shown dimmed on the map with their ore visible) and the game opens a second, random one. Contents are set by distance from the core so new resources arrive slowly. Enemies enter over any edge of the opened land. This replaced the resource-module pick.	BUILT
Ranged enemies and tougher health	Enemy health is about 35 percent higher and a growing share of each wave shoots from range, so a defence of only turrets is no longer enough: walls, robots and coils matter. Numbers are a first pass and need play-testing.	BUILT
No storage chests	Chests were removed. The core stockpile is the only store; belts carry everything else.	BUILT
Pausing	While paused the screen is dimmed with a PAUSED overlay, and only camera movement and zoom work. Nothing can be built, erased, moved or started.	BUILT

TOPIC	DECISION	STATUS
Endless mode	Currently open to everyone: the levels simply continue past level 30. Whether it should wait for three stars is still to decide.	BUILT

STILL OPEN

- NEW DIRECTION (proposed by the producer, not built): the player is an invading force that has landed on an alien planet, and the enemy is organic (creatures, swarms, hives) instead of rogue machines. Later, each run could land on a random planet type with its own enemy set. This changes the story, the enemy art and names, the tagline and the ground look.
- Is a 24x24 starting square (the map is now 200x200) big enough for level 1? Needs play-testing.
- Are the boons and omens exciting without being too generous? How often should a boon be offered (currently every second level)?
- Should enemies be able to spawn from every open edge, or only from the far side of the explored land?
- How should commander bonuses change now that the module choice has gone (Tamsin had an extra module pick)?
- How long should a build phase be, and what should starting a fight early reward?
- How big a role should research play, when the run is only a few hours at most?
- Multiplayer or shared leaderboards? Worth a server, or keep the game fully offline and free to host?
- Monetisation, if any: free on the web, paid on Steam, cosmetic extras?

WHO WOULD HELP MOST

Balance designer

Someone who enjoys spreadsheets and playtesting to tune enemy waves, prices and progression so levels 10, 20 and 30 feel earned.

Sound designer / composer

The game is silent today. It needs a dark industrial soundtrack, machine loops, weapon and lightning effects, and an audio identity for the FuzeNova intro.

Artist / animator

The core art is AI-assisted and consistent, but boss animations, walk cycles, tile transitions and UI polish would lift it a lot.

Web developer

For a global leaderboard, cloud saves and account features, kept separate from the site's existing database.

Playtesters

People who play Factorio, Mindustry or Satisfactory and will say honestly where it is confusing or dull.

Producer / marketer

Trailers, an itch.io and Steam presence, community, and the legal checks around names.

LEGAL NOTES

- The title, commander names and enemy names are original but unchecked. A trademark search (UK IPO, USPTO, Steam, itch.io) is needed before release. An earlier working title was dropped because it already exists on Steam.

- Mechanics inspired by Factorio and similar games are not protected, but their art, names, sounds and specific designs are. Everything here is original art. "Tesla coil" is a generic term, and this game calls its version the **Storm coil**.
- Art is AI-assisted (Gemini). Check the current terms on commercial use and disclosure for any store you publish to.

19 Change log

2026-09-30

- Share card: a 1200x630 preview image (og-image.png) and the Open Graph / Twitter tags, so links to the game show the logo and a battle scene on social media and chat apps.
- Weapon families and damage types: every weapon now goes base -> one of two level-1 types -> one level-2 weapon, upgraded in place by double-clicking. Projectile: Gun turret -> Scatter gun / Sniper -> Artillery (range 4-16, bursts over an area, fires artillery shells). Fire (new, researched with Flame designs): Flamer -> Incendiary launcher / Focused torch -> Plasma cannon, fuelled by coal, charcoal and wood. Lightning: Storm coil -> Shield coil (halves damage to nearby buildings) / Stun coil (freezes enemies for a moment) -> Railgun (long piercing energy shot), researched with Coil designs. A level-2 weapon needs a level-1 type first (either), and going back down is free. Four damage types (kinetic, flame, energy, lightning): flat armour only stops kinetic, and every enemy is weak to some types and resists others (shown on the Enemies tab). Enemies on fire keep burning; the Marshal Ozric and Cinder Queen Ysolde commanders, who were waiting for these weapons, now play (unlocked by having 3 Artillery or 4 Flamers standing).
- Embers and the Workshop (out-of-game progression): every run now earns Embers for the levels it got through (1 each for the first three levels, 2 for the next three and so on; Easy pays 0.75x, Hard 1.25x, Extreme 1.5x), whether it ends in a destroyed Core or is left behind by starting a new game, because the game is endless. Embers are spent in the Workshop (a button on the title screen) on ten permanent upgrades with many small levels: turret damage, ammunition, wall and turret health, Core health, robot health, robot damage, faster robot building, coil damage, a bigger starting stock, and one extra map square to choose. Progress is saved to the player's account and merged between devices.
- Robot production chain: four buildings now make robots from plates, each step up needing another kind of plate. Drone workshop (copper + iron: Scout drone, Scout walker), Gunship hangar (+ tin: Gunship drone and the new Sapper drone, which bombs a patch of ground enemies but cannot hit flyers), Walker foundry (+ lead: Trooper, Turret walker) and Heavy works (+ tin and lead: Heavy walker, Mobile artillery, Titan and the new flying Carrier, which launches and keeps up to four small drones). Every building holds 12 space of robots, so a Heavy works holds four Heavy walkers (now 3 space), or two Artillery, or one Titan or Carrier. Buildings are paid for in iron and copper from the Core; the tin and lead are belted in as ingredients. Old fabricators become Drone workshops and keep their plates. The buildings use the existing launch-bay, foundry and titan-forge pictures; the Sapper drone and Carrier have their own new top-down art.
- Instructions brought up to date: the How-to-play guide, the Buildings and Robots tabs, the help cards, the tutorial, the inspect panel, the controls list and the new-item pop-ups now describe the robot chain, tin and lead as ingredients, turret upgrades, underground belts and the scrap bin.
- Underground belts (the Tunnel, key U, 8 iron plates, free to remove): a pair of 1x1 pieces facing the same way that carry items under walls, machines, rocks and other belts. Put an entrance at the end of a belt and a second piece in line in front of it with up to 4 tiles in between; the second becomes the exit by itself. The pieces are solid, so enemies cannot use them to get through a wall line, and the tiles in between stay free to build on.
- Turrets fed from the same belt now share it out: a turret does not take an item if a turret further down the same belt (counting what is already on its way) is emptier, so the first turret no longer fills up before the others get any. Power poles are tougher (140 health, was 40; generators 220) and enemies only notice poles and generators within 4.5 tiles, after walls, turrets and coils.
- Robots can no longer be left stuck between walls and buildings: one that is chasing an enemy but not moving is now noticed (before, only idle ones were), and if it is walled into a small pocket with no way out it is lifted to the

nearest free ground outside after about 3 seconds. One-tile and two-tile corridors between walls are walked straight through.

- The Titan no longer scrapes along scenery: big robots plan their routes keeping a tile clear of rocks, trees, pools, walls and machines, so they take a wide gap in preference to a one-tile squeeze (and still squeeze through if that is the only way).
- Instructions: the How-to-play guide, the Controls list and the Research-complete message now explain turret upgrades (double-click a gun turret), the scrap bin, and how to read the status of an assembler.
- Enemies now go for power poles and Ember generators as well: they count as targets within 7 tiles like walls, turrets and Storm coils, but 2.5 tiles farther away, so a wall, turret or coil that is nearly as close is still hit first. Ranged enemies shoot them too.
- Solid buildings (walls, turrets, machines, poles) can no longer be placed on top of a walking enemy; the ghost turns red until it has moved. Belts, inserters and crossovers still can, and flyers never block building.
- The Buildings tab of the menu now shows the range of the gun turret and Storm coil, and has cards for the Scatter gun and Sniper upgrades (range, fire rate, damage, cost and how to unlock them). The Robots and Enemies tabs already list the range of each unit.
- Power lines show their state: a network with full power has glowing blue wires with charges running along them and a blue lamp on each pole; one that is short of power turns amber with slower pulses; one with no power (no burning generator) is dull grey and dashed.
- Robot fabricators now have room for 12 space of robots (was 10), so mixes of 2-, 4- and 6-space robots fill exactly. The Turret walker now takes 6 space (was 5) and the Titan takes the whole fabricator, 12 (was 10). A small ding plays when a wave is cleared (not after the last one, which has the level-complete fanfare).
- Staying signed in: the sign-in is kept on the device until the player signs out (set explicitly, and the browser is asked not to clear it). With no connection, a device that already has a confirmed account goes straight to the title screen instead of the sign-in form.
- Scrap bin (new building, 2x2, 20 iron plates, key B): destroys any item an inserter or belt gives it, so leftovers no longer jam a belt. Put a filtered inserter in front of it to remove only one kind of item. It has idle and burning pictures.
- Discovery pop-ups: the first time a kind of item turns up in the factory (gunpowder, bullets, steel and so on) the game pauses with a card explaining what it is for, until the player clicks Got it (or presses Enter). What has been seen is remembered per player and saved to their account; an existing player is not shown pop-ups for items they already have. Not shown in the tutorial.
- Robots answer an attack anywhere in the base: with no enemy within 16 tiles they march to the nearest enemy, but only one within 30 tiles of the fabricator that built them (or the Core if it is gone), so they never wander off across the map.
- Assembler status: each assembler on the map now shows what it holds of every input (have / needed, red when short), a badge saying what it needs, and a "N ready - take out" badge when its output is full (6) and it has stopped until an inserter takes the packs out. The inspect text is labelled, and the research panel shows how much of each pack is in the Core against the cost ("need N more") and which pack kinds have an assembler making them.
- Map squares after a battle: the squares offered, and the bonus square the map opens, are now always the ones closest to the core (random only between squares equally near), so the explored land grows outward without big gaps.
- Bosses can no longer be trapped by scenery: a boss that has not moved half a tile in 3 seconds (and is not chewing a building) crushes the rocks and trees in its way towards the core; if that is not enough it is set down on the nearest ground from which the core can be reached.

- Update prompt: a running game (browser tab or installed app) now checks for a new version every 10 minutes and whenever the player returns to the window (one tiny request), and shows a "new version is ready - click to reload" button when one is live.
- Ground patches: churned-mud decals lie over mud flats, and scorched cracked ground (some with glowing lava cracks, one crater), ash dunes and salt flats are scattered thinly over open ground as flat, non-blocking decoration, kept clear of ore, rocks, buildings and the core.
- Terrain features: mud flats (soft brown ground that slows anything on foot to about half speed, enemies and robots alike, but you can build on it; enemies prefer to go round it if the detour is short) and toxic pools (green pools that block walking and building, like tar). Both are drawn as smooth organic blobs. Mud never covers ore or rock and stays well away from the core. Flyers are not slowed.
- Walls now show their health: every wall piece (straight, corner, T, cross and the lone block) has a cracked version below 66 percent health and a broken, smoking one below 33 percent. The Gatling brute and the Laser spider show their insulation (white ceramic discs, earthing straps), and the Siege crawler and the Acid cannon show their armour plating.
- Art pass: the twelve boons and nine omens have painted icons (boon cards; an omen chip with the icon and a one-line description in the top bar); twelve HUD icons (core health, kills, robots, power, cooldown, call-wave horn, research, armour, insulated, ranged...) replace words in the top bar, the cooldown banner, the buttons and the enemy cards; eleven pieces of scenery (bones, a wreck, scrap, fungus, a crater, a buried arm, ash, a pipe, pebbles, a shrub) are scattered over open ground, the same for every seed, clear of ore, rocks and the core; and destroyed buildings leave a heap of rubble that fades after about 45 seconds.
- Crossover and splitter now have painted art: a arrow-free crossover tile (the belts can cross either way) with a riveted plate and corner lamps, and a splitter gate with a gear and divider between its two lanes. A `HANDOVER.md` file in the game folder explains how to run, test, build and deploy the game, the code map, the rules we settled on, and what is left to do.
- New art in the game: the Scatter gun (orange hazard-striped pedestal, flared double barrel) and the Sniper (cyan-lit pedestal, long scoped barrel) have their own sprites, and the four science packs (projectile orange, electromagnetic blue, robotics green, advanced purple) have proper glass-flask icons shown in the stock and research panels.
- Trees redrawn at a sensible size: they are scaled by their longest side so none is more than about two tiles tall (the thin ones used to be stretched to five), and the two bare-pole trees that looked like sticks are no longer used.
- Research is harder to rush: packs are spent at once but a research then takes factory time to finish (15 seconds for the first level, 10 more for each level after, counted only while the factory runs: fights and the cooldown), one at a time. Level 1 of a tech needs only its own pack; level 2 also needs a second kind of pack (a second supply chain); level 3 onwards also needs the new Advanced science pack (steel, bullets and bronze: the whole ammunition chain). The later levels also need the run itself to have reached level 3, 6, 10 and 15. The research panel shows the countdown, the gate and the cost.
- Armour: heavier enemies (armoured and siege crawlers, gatling brutes, laser spiders, cannons, siege spitters, bombers and the bosses) take a flat amount off every hit (never more than three quarters of it), so iron-plate shots and the later hops of lightning barely hurt them and bullets matter. Gatling brutes and laser spiders are also insulated: Storm coils do only 40 percent to them. The menu shows each enemy armour.
- Double-click pickers: an Assembler opens a grid of the recipes it can make (locked ones say what unlocks them), and a Robot fabricator opens a menu of robots with health, damage, range, speed, room, cost and build time. Single clicks no longer cycle through them.
- Turret branches: researching Turret designs lets you upgrade a gun turret in place (double-click it) into a Scatter gun (short range, a wide cone that hits every enemy in it) or a Sniper (range 12.5, slow, five times the damage, shrugs off most armour). Upgrades cost plates, keep the turret ammunition, and are free to switch back.

Upgraded turrets wear a coloured ring (orange or blue) and a different barrel; proper art for them is still to come.

- Waves lean towards your weak side: the first wave of a level can come from any open edge, but later waves are weighted towards the edges with the least defence nearby (turrets and coils count most, then robots, then walls), so a defence that only covers one side is tested on the others.
- Enemy pathfinding rebuilt as a flow field: one search from the core outward gives every tile its cost to the core, and ground enemies simply step to the cheapest neighbouring tile, instead of one route search per enemy. Rock and unexplored ground are impassable; belts and inserters cost nothing; every other building is passable at a high cost, so enemies go round a wall when there is a gap but chew through the cheapest wall if the core is sealed in. Big enemies (siege crawlers, brutes, bosses) use a second field that keeps a tile clear of rock, so they do not squeeze through one-tile gaps; the Siege Brute smashes through buildings and is stopped only by terrain. The field is rebuilt when buildings change, at most four times a second. Enemies now steer apart so a wave spreads out instead of stacking on one tile, spawn points are moved out of rock or sealed pockets, and an enemy that has moved under half a tile in 3 seconds is pushed along the route, then set down nearer the core, then given up on. 200 enemies cost about 3 ms per step.
- The core now only accepts what can be spent: iron plates, copper plates and science packs. Raw ore, ammunition, tin and lead plates and the rest are refused, so a belt or inserter holding them waits at the end and a warning mark says the core refuses it. This stops ore (500 of it, for example) piling up in the core doing nothing.
- Cloud saves are lighter on the account service: a save is uploaded at most once every 30 seconds, never when nothing has changed since the last upload, and always when the tab closes or the player signs out. A new version of the game now reloads itself on the title screen, or offers a reload button during play.
- Call the next wave early: during a breather a green button (or the C key) skips the rest of the calm and sends the next wave at once, paying iron plates for the seconds skipped (a little more on later levels). It plays the warning horn. Not available while enemies are alive, before the first wave or after the last.
- Waves with a guaranteed breather: the next wave never starts until the last has been destroyed and a calm of about 25 seconds at level 1 (down to 14 by level 30) has passed; only a wave that drags on for 90 seconds is not waited for. Waves also build through a level: the first is about 60 percent of an even share and the last about 140 percent, so the pressure climbs to a finale. The top bar shows the wave number and, during the calm, a countdown to the next wave.
- Placing or moving a building onto a ground enemy or robot now pushes it out to the nearest free ground instead of trapping it inside (flyers and flat belts and inserters are unaffected). Any enemy found inside rock is also lifted out.
- Storm coils recharge much more slowly: a coil gained 60 charge a second but a bolt costs 50 every 1.3 seconds, so it never ran down and fired non-stop. It now gains 20 a second while still drawing the same power, so a full coil fires a burst of four bolts and then about one every 2.5 seconds. More coils, more generators or a long gap between fights are how you get more lightning.
- Live attack glow on the minimap: wherever a building, a robot or the core is being hit, a hot orange bloom pulses on the minimap and fades about a second and a half after the hits stop (nearby hits merge into one glow). Enemies are red dots and new spawns still ring.
- Fixed a stuck ring: the spawn and straggler pings only faded during a fight, so one left over when the level ended stayed on screen as a motionless circle until the next fight. They now fade in every phase and are cleared when a level is won.
- Sound tab in the game menu with separate Master, Music and Sound effects volume sliders (effects include the machine hum) and a mute box. They change what is playing at once, play a sample when you let go of the effects slider, and are remembered on the device.

- Email verification: a new account is sent a confirmation link and cannot play until it has been clicked (existing accounts are asked the first time). Throwaway-mail domains are refused at sign-up, and the cloud-save rules only accept verified accounts.
- Fights rebalanced: waves now wait until the last wave is dead plus a breather (24 seconds at level 1, down to 12) before the next starts, so there is always a real gap; counts and strength were dialled back (about 30 percent more enemies than the original, packs of 2 to 14) and the tougher types (siege crawlers, blade spiders, sprayers, gatling brutes, cannons) arrive later, so difficulty climbs steadily. Headless test: 2, 2, 5, 5, 7 turrets at levels 1, 3, 5, 6, 8.
- Enemies caught on rocks: an enemy that is not fighting anything and makes no headway for 10 seconds is set down on free ground nearer the core; if it gets caught again it gives up and falls. Together with the straggler rule this means a fight can never hang on scenery.
- Tar pits redrawn as smooth, glossy black pools with a crusted, ember-flecked lip and a dark stain on the ground round them, instead of square tiles with visible seams.
- Moving a building (V) now empties it: a moved turret arrives with no ammo, a smelter or assembler with no items, a generator with no fuel, a coil with no charge. Settings (recipe, filter) and health are kept. This closes the loophole where a turret could be carried across the base mid-fight and keep firing without being supplied.
- Accounts and cloud saves: players must sign in with email and password (or create an account with a username) before the title screen. The game uses the same Firebase project as Crystalbound Saga, so one login works in both. The saved run and the progress (unlocked commanders, stars, last choices) are stored in the player cloud document (collection cinderSaves, one document per player, readable and writable only by that player) and follow them to any device; signing out clears the browser copy. Needs the new rules in firestore.rules published in the Firebase console.
- The cooldown is now impossible to miss: a green COOLDOWN banner with a countdown and a draining bar appears in the round panel, and a message flashes at the start. The victory music plays while you choose your boon and square, then the build track and the running factory take over for the cooldown.
- Stragglers: once everything has spawned and only 1 to 3 ordinary enemies are left, they are pinged on the map after 10 seconds, run twice as fast after 18, and give up after 35, so a fight can never hang on one lost enemy.
- Music follows the quiet: in the breather after a win the calm build track plays (it already did) and the machines are heard again; in the lull between waves of a fight the calm track fades in after two clear seconds and the battle track returns when the next wave starts. Boss fights keep the boss track.
- Breather after each win: the first part of the build phase is a cooldown in which the factory and robot fabricators keep running (50 seconds on Easy, 30 on Normal, 20 on Hard, 12 on Extreme at level 1, shrinking as levels pass), shown in the top bar. After it the factory pauses for the rest of the build time. This balances the faster wave fights with time to mine and build.
- Waves: a fight is now split into 4 to 9 waves (more on higher levels). Each wave is a quick surge: its packs arrive about a second apart, mostly from one side, then a short breather of roughly 15 to 28 seconds to mine and repair before the next wave. The next wave comes early if the last one is nearly cleared. There are 60 percent more enemies than before in packs twice as big, each a bit weaker, so the field is busy and the fight is long but the total strength is only modestly higher. Measured with the headless test: 2, 4 and 6 turrets needed at levels 1, 3 and 5.
- Belts, inserters, crossovers and splitters are now always free to build as well as to remove, so laying out a factory never costs plates. Everything else still costs plates.
- Forests no longer appear next door: wood is only a weak generator fuel, so offering it on the first choice wasted a pick. Next-door squares now hold only iron and copper; forests start two squares out.
- Combat sounds added and wired: coil zap, robot shot, bullet hit, enemy death, boss arrival, structure hit, structure destroyed and a core-damage alarm (only when the core is below 35 percent). Sound effects are

louder against the music. Every menu, panel, button, tab and card restyled in one modern glass-and-ember look. The first iron patch is bigger and deeper (about 45 to 75 tiles).

- Pause: pressing P dims the whole screen with a PAUSED overlay. While paused nothing can be built, erased, moved, researched or started; only the camera works. The title screen no longer shows the map behind the logo.
- Commander portraits added (all nine) and the new-game screen rebuilt as a carousel: arrows, dots, keyboard, swipe, portraits in arched frames, locked commanders as silhouettes, difficulty as four tiles with flames and stat chips, and a big Start button.
- Enemies: base health up about 35 percent; ranged enemies appear earlier and more often, and the laser spider now shoots from range. Boons are weaker (about half) and offered every second level; the instant rewards are smaller.
- Map expansion: the map is a grid of 24x24 squares (A-G, 1-7). A run starts with only the square round the core, which holds a small iron patch and nothing else. After each win you choose one of three neighbouring squares to open and the game opens a second, random one; enemies come in over the edge of the opened land. What is in a square depends on its distance from the core: copper next door, then coal and tin, then lead and sulfur, then the far forests. The first time a new ore appears, a message explains what it is for. This replaces the resource-module choice. The music no longer restarts when a button is clicked.
- Roguelite layer: every run has a random map seed (ore patches, ammo-chain resources, rocks, trees and tar all move; the map number is shown in the HUD and ?seed=N replays one). From level 2 each fight may carry an OMEN shown before it starts (Swarm, Ironhide, Swift, Acid rain, Two fronts, Bounty, Brutal, Ash fog, A lull). Every win offers a choice of three random BOONS (turret damage/rate/range, coil damage, extra lightning hop, robot health/damage, sturdier defences, or instant windfalls: plates, science packs, free scouts, a full repair), then the resource module. Starting stock and level rewards were trimmed so the economy snowballs less.
- Storage chests removed (old saves put their contents in the core). Ranged enemies added: acid spitters, acid sprayers, gatling brutes, acid cannons, siege spitters and combat drones stop short and shoot, and some out-range a gun turret. The 3x2 drill keeps the plate-and-hopper look.
- Inserter filters: click an inserter to choose the one kind of item it moves (default: any). A filtered inserter shows the item on its corner.
- Storm coils now store energy: they charge up to 200, each bolt spends 50, and a full coil draws almost no power. Robot fabricators have room for 10 space of robots each (scout 1 ... Titan 10), shown as a gauge. New art in the game: rocks, burnt trees, tar pits, and the 3x2 drill.
- Research is now paid in science packs you craft (projectile, electromagnetic, robotics) and it unlocks robot designs and ammunition. Levels end as soon as the last enemy dies, no waiting for the clock. Waves are bigger and tougher. Robots no longer heal between levels. Spawn pings on the map, edge arrows and the minimap. Fast-forward (F, 1x/2x/4x).
- Smarter robots: idle robots go to their own guard post just outside the base (spread evenly round it) and face outward instead of circling through the buildings; they push gently apart so they never pile up; they no longer clip building corners; and a robot that stops making headway re-plans, then is lifted onto free ground.
- Mining drills are now 3 long and 2 across (turn with R). Ore leaves only from the middle tile of each long side, so the belt always starts in the middle of the drill. Drills from older saves keep their old 2x2 shape. Cost 16 iron.
- Smaller first map: the playable square starts at about 52 tiles across and grows with each level; outside it is unexplored and unbuildable. Natural obstacles added (boulder fields, burnt trees, tar pits and rock ridges with gaps) that block building and block ground units. Enemy ground units now path round rocks and buildings, and chew through whatever blocks them.
- Starter commander added (no bonus, no drawback); the six others are unlocked by achievements. Belt crossovers and splitter/mergers added. Minimap added. Range circles show on hover and while placing turrets, coils and poles. Range research added (turrets, coils, robots).

- Tester pack: npm run tester builds release/CinderAutomata-Tester.zip, a Windows double-click launcher (Play Cinder Automata.bat) with a small built-in local server, plus a Mac/Linux script and a read-me for testers.
- Belts now always connect visually: the first tile of a line bends away from an inserter that feeds it, and dead-end tiles turn to face what they feed, whether built before or after it, and in existing games.
- Ground robots now path round buildings instead of walking over them (belts and inserters are walkable); flying robots are unaffected. Belt ends turn to face whatever they feed (inserter, core, turret, fabricator, smelter...). Ore on belts is larger and scattered across the belt width so a full belt looks like a heap.
- Machine and ambience loops added (drill, smelter, belt, assembler, generator, coil hum, wasteland wind, battlefield bed). All set to the same loudness; the game plays them by what is on screen while a fight runs.
- Building sounds added (place, belt, remove, rotate, pick up, put down; several takes each, picked at random). Music: a calmer track now plays while building; every fight opens with the battle track and, on boss levels, crossfades to the boss track once the boss is in sight. Star sound replaced.
- Boss music (Relentless March), victory music (Triumph in Steel) and defeat music (Silent Descent) added and wired: the boss track plays on boss levels; the victory and defeat pieces play once when a level is won or the core falls.
- Difficulty levels (Easy, Normal, Hard, Extreme) and achievement-based commander unlocks added, as agreed with collaborators. Fight music for levels 1 to 9 (Front Line Defense) added.
- Build-phase music (Factory Dawn) added. Music now follows the phase: build track while building, fight tracks by level range (fight-1, fight-2, fight-3, boss) once they exist.
- Menu music (Ember Glow) added: plays on the title screen and commander select as a seamless loop and fades out when a game starts.
- Colossus Walker art replaced with a new version. Decisions recorded: difficulty levels, commander unlocks through achievements, structure-repair cost by difficulty.
- Sound effects started: interface sounds are in and played (click, menu open, refusal, fight start, level complete, star, defeat, build phase). Sound on/off button (N).
- Research added (press T): projectile damage and rate of fire, electromagnetic damage and extra lightning hops, robot plating and weapons, reinforced structures. Base turrets fire a little slower (0.42s) and Storm coils start weaker (30 damage, 2 hops) so research has room to matter.
- Inspect mode (F3): hover anything to see what it needs and how to connect it. Buildings placed in a build phase are blueprints (blue, dashed) that lock in when the fight starts; removing a blueprint is free. Belt ends beside the Core now turn to face it.
- Removing belts and inserters now refunds them in full; other buildings still refund 75%. New: move a building for free with V; F2 controls panel; warning marks over mis-wired machines.
- Commanders added: a select screen with six playable commanders (bonus and drawback each), stars for completing level 10, 20 and 30, and Endless play past level 30 with best level recorded. Artillery and Fire commanders are shown as coming soon.
- Bosses added: Siege Brute (level 10), Colossus Walker (level 20), Hive Queen (level 30), repeating every ten levels. Each walks in 30% of the way through the fight, the level cannot end until it is dead, and it pays a bonus. Health bar on the map and boss name in the top bar.
- Balance pass 1: enemies now arrive in packs and health scales faster early, slower late. Measured with a headless test (tools/balance.ts): about 9 turrets with endless ammo hold level 10, about 16 hold level 18.
- First version of this document.
- Storm coil, generators and power poles added, with painted art and lightning effects.
- Menu with a picture guide of every building, robot, enemy and recipe. Interface rebuilt to scale with the window.
- Self-connecting walls. Logo added to the title screen. Installable web app (offline play) added.

- Building costs, core stockpile, refunds. Save and Continue.
- Assemblers, ammo chain (bullets, shells), resource modules after each level.
- Robots redone as top-down sprites that turn to face targets; enemy waves with 10 enemy types.

This document is generated from the game's own code and a text file of notes (docs/gdd/content.ts). Run `npm run gdd` to rebuild it after any change.